

Sidharth Parameswaran R.

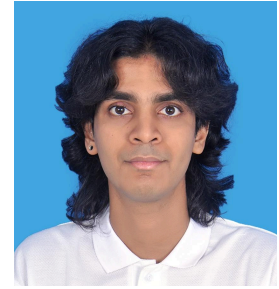
Mechatronics Graduate - Data Engineer

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EDUCATION

FH Upper Austria, Wels, Austria — M.Sc Automotive Mechatronics and Management

SEP 2022 - MAR 2025

- **Master's Thesis:** Data Intelligence Applications for the Validation of On-Board Emissions Monitoring.
- Developed statistical data pipelines in Python to validate telemetry signals from vehicle sensors against Euro-7 regulatory thresholds.

VIT University, Chennai, India — B.Sc Mechanical Engineering

JUL 2017 - JUN 2021

- **Bachelor's Thesis:** Performance, Emission and Combustion Characteristics of a Biogas–Diesel Dual-Fuel Engine using Taguchi Method.

PUBLICATIONS

Student Centric Feedback Based Case Study on Modern Educational Methodologies in Engineering Subjects – can we Fill the Industrial and Educational MINT Gap?

[DOI](#)

- This paper was written in collaboration with the professor-mentor and other members of the IEEE Student Wels branch in the summer of 2024.
- It was based on the experiences of the students in a hackathon that took place in Gran Canaria, Spain.
- The main part of this paper is a student-feedback based evaluation of the hackathon concept.

Performance, emission and combustion characteristics of a biogas–diesel dual fuel engine using Taguchi method

[DOI](#)

- This paper analyses the usage of a biogas–diesel mixture in the place of traditional fuels in terms of performance, emissions and combustion characteristics then exhibited by the engine in question.
- This study was initially performed as part of a bachelor's thesis and was then adapted into a research paper with the help of other collaborators. Published in the Materials Today journal.

Core Competencies

Software Development

Built and maintained production Python tooling and Java features in enterprise codebases. Foundation in object-oriented design, version control, and AI-assisted development workflows. Active personal projects in Linux systems.

Data Analysis & Modelling

Python (Pandas, NumPy), SQL, ERP/enterprise data flows; basic understanding of data pipeline architecture. Skilled at presenting data analysis results.

Modelling & Simulation

Modelling of automotive components and machinery. Simulation in the commercial vehicle sector.

EV Technologies

Experience in modelling, simulation of EV charging activity. Processing, analysis of charging, energy data and modelling how external factors such as weather can affect it.

Presentation and Communication

Skilled at extracting and presenting necessary information in both an academic and industrial context.

Multi-disciplinary Background

Strong foundation in mechanical engineering concepts such as mechanics, hydraulics. Supported with experience in mechatronics, programming, testing and software dev.

WORK EXPERIENCE

AVL List GmbH, Graz, Austria — Werkstudent in Data Intelligence for Engine Calibration

AUG 2023 - NOV 2024

- Development of a cloud-based diagnostics platform, written in Python, to replace legacy tooling that was used in automotive calibration.
- The automation of previously manual workflows reduced processing and data visualisation time by up to 70%.
- All the findings from this industrial work that were related to the Euro-7 regulations were incorporated into a master's thesis which was submitted and defended in March 2025.

Kreisel Electric GmbH, Rainbach im Mühlkreis, Austria — Project Intern

MAR 2023 - JUN 2023

- Worked on the simulation and optimisation of the company's high-powered EV charger model using Modelica.
- Implemented a model that simulated changes in charging curves based on weather data.

Cognizant, Chennai, India — Programmer Analyst

FEB 2021 - JUN 2022

- Maintained a legacy DBMS tool written in Java and SQL within a production environment. Worked with Git, CI/CD as a part of an international team.
- Implemented new sorting features, executed stringent tests and deployed.

SELECTED ACHIEVEMENTS

Europe Tech Hackathon 2026, Vienna, Austria

MAY 2026

- Designed a working prototype that connected sample Digital Product Passport data to an ERP system. Worked with Python, JSON files, and API keys to establish backend connection.
- Awarded **Best European Idea**, one of the four awards presented at the event.

Smart Green Island Makeathon, Gran Canaria, Spain

MAR 2024

- Contributed to a cross-national team developing a sustainability prototype under the LEONARDO project.

IEEE Student Secretary, Wels, Austria

MAR 2023 - APR 2024

- Co-led the re-establishment of the FH Upper Austria IEEE student branch from inactive status.
- Organised and ran three technical and networking events. Networked with other branches in the country.

Technical Skills

Programming: Python, C, C++, Java, SQL, Git, Bash, APIs. Databases.

Software: Jira, MS Office Suite, Linux.

Engineering: MATLAB Simulink, Solidworks.

Documentation: LaTeX.

Languages

English - Native.

Hindi - Native.

Tamil - Native.

German - A2.